



AnalogLink Sentinel: Non-Invasive Digitization of Analog Pressure Gauges via AIoT for Real-Time Monitoring

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ABSTRACT

The transition toward Industry 4.0 is significantly hindered by the prevalence of legacy analog infrastructure lacking connectivity capabilities. This paper presents AnalogLink Sentinel, an AIoT solution designed for the non-invasive digitization of analog pressure gauges. The system utilizes the vision sensor of the Seeed Studio XIAO ESP32-S3 Sense to capture gauge imagery, which is processed locally using an image classifier trained on the Edge Impulse platform. Unlike traditional geometric regression methods, this model visually categorizes the needle's status into specific pressure ranges in real-time. The architecture integrates Wi-Fi connectivity for telemetry transmission to Ubidots, local visualization via an OLED display, and physical actuation through relays for alarm management and pump control.

I. INTRODUCTION Y JUSTIFICATION

The digital era and the industry 4.0 paradigm face a critical bottleneck: the prevalence of legacy analog infrastructure lacking connectivity, communication, or data transmission capabilities. In Colombia, diverse sectors such as agribusiness, power generation, water treatment, and food processing rely on robust machinery with operational lifespans spanning several decades. Nevertheless, the management of these industrial processes still depends on manual visual inspection, which fragments information flows and creates data silos. The emergence of AIoT enables bridging this gap by integrating advanced perception sensors into embedded systems. AnalogLink Sentinel is presented as a response to the need for the non-invasive digitization of analog instrumentation using lightweight neural networks. This approach not only modernizes existing hardware but also democratizes access to real-time data and analytics, allowing enterprises to achieve a comprehensive technological overhaul at an affordable cost.

AnalogLink Sentinel is built upon three fundamental pillars:

- **Risk Mitigation:** Manual data acquisition from analog pressure gauges is a time-intensive process that requires operators to travel to the physical site and introduces significant error variability due to fatigue or human misreading. Consequently, the system eliminates temporal gaps by providing immediate responses through automated alerts whenever a critical pressure level or operational anomaly is detected.
- **Non-Invasive Architecture:** The installation of conventional digital pressure transducers typically entails pipe depressurization, welding, specialized technical certifications, and even costly plant shutdowns. AnalogLink Sentinel offers an external, non-intrusive solution that does not directly impact the process flow, thereby enabling the modernization of control strategies without compromising industrial productivity.
- **Energy and Maintenance Efficiency:** The implementation of this system extends beyond simple needle reading; it facilitates the development of predictive maintenance models using real-time digitized historical data. Furthermore, it enables the optimization of pumping system control loops based on real demand captured via AI, leading to a significant reduction in electrical energy consumption.

Consequently, AnalogLink Sentinel functions not merely as a digitized pressure transducer, but as a proactive monitoring framework that integrates perception, connectivity, and autonomous actuation to optimize operational efficiency and safety in critical industrial environments.



Recent literature has documented the implementation of analog gauge digitization systems through AIoT, showcasing a significant trend toward Edge Computing architectures that perform local data processing. Researchers such as Katsigiannis et al. and Peixoto et al. have demonstrated the feasibility of utilizing low-cost development boards—such as Raspberry Pi or ESP32-CAM microcontrollers—directly integrated into measuring instruments to capture and wirelessly transmit imagery. [1], [2]. Regarding the inference and readout phases, the literature emphasizes the integration of computer vision techniques with artificial intelligence algorithms. Studies by Yang et al. and Chu et al.

deploy neural networks for instance segmentation and keypoint detection (leveraging architectures such as YOLOv8, YolactEdge, or RTMPose) which are executed on edge devices like the Jetson Xavier NX or high-performance GPU servers [3], [4]. Furthermore, the literature reports the utilization of geometric rectification methods—including the Hough transform and polar projections—achieving high precision with mean errors below 1%. In the field of industrial automation, systems such as those proposed by Corrales et al. validate the direct integration of these digital readouts into industrial control loops to drive physical actuators. [5].

TABLE I

COMPARATIVE ANALYSIS OF ANALOG INSTRUMENTATION DIGITIZATION SYSTEMS BASED ON AIoT AND COMPUTER VISION ARCHITECTURES

Sensors	Actuators	Hardware	AI Techniques Employed	Achieved Evaluation Metrics	Ref.
Logitech C920 HD Pro Webcam	Control loop integration for device actuation upon set-point deviations	Raspberry Pi 3 embedded platform	Lightweight Artificial Neural Networks (ANNs), Gaussian Filter, Canny edge detector	MAE: 0.589 PSI, R ² : 0.985, Cycle time: 3.74 ms	[5]
Camera sensor coupled with a 3D-printed enclosure	Triggering of alarms and warnings configurable via an IoT web platform	ESP32-CAM Microcontroller	Hough Transform, HSV mask segmentation, polar transformation, and pixel projection	Average Relative Error: 0.95%, Max Absolute Test Error: 0.217 Bar	[2]
20-MP onboard drone camera	N/A (Focused strictly on passive inspection/telemetry)	DJI Mavic 2 Pro UAVs, Server with NVIDIA GTX 3090 GPU	CSPNet architecture, YOLOv8 detection, RTMPose localization, Graph Attention Units (GAUs)	Detection Precision (AP50): 99.4%, Readout Accuracy: 85.5%, Speed: 30.9 FPS	[4]
Industrial-grade plant cameras	Real-time stress status and early warning system	NVIDIA Jetson Xavier NX Edge computing device	SSD (VGG16), YolactEdge segmentation, Elliptical affine transformations	Mean Reference Error: 0.0237%, Average Relative Error: 0.0300%, 10.26 FPS (INT8)	[3]
Digital camera	N/A (Used for digital evidence generation in preventive maintenance)	Smartphones	Preprocessing with Nick binarization and Convex Hull, EfficientNet-B0 CNN, YOLOv6 head	Detection F1-score: 89.13%, Mean Angular Error: 4.11°, Relative Error: 0.0131	[6]

II. PROPOSED SOLUTION

The AnalogLink Sentinel system is conceptualized as an intelligent monitoring unit designed to bridge the digital gap in industrial analog infrastructure through a strictly non-invasive approach. This technical proposal enables the modernization of legacy instruments without direct intervention in production processes or the requirement of

plant shutdowns, thereby facilitating real-time data acquisition within Industry 4.0-aligned environments.

At the device's core, the vision sensor of the Seeed Studio XIAO ESP32-S3 Sense embedded system handles the acquisition of visual data from the pressure gauge. Data processing is performed entirely at the edge utilizing an image classification model developed on the Edge Impulse



platform. This model is designed to visually categorize the needle's status into specific pressure ranges, thereby optimizing response latency and ensuring a robust deployment on the microcontroller by focusing on the identification of discrete pressure states.

To ensure effective user feedback, the solution implements a three-tier architecture encompassing local visualization, cloud-based telemetry, and physical actuation. Inference results are displayed in real-time on an in-situ OLED screen and are simultaneously transmitted via Wi-Fi to the Ubidots platform for reporting and historical data logging. Ultimately, the system closes the control loop through relay switching, triggering audible alarms or activating backup pumps when pressure levels deviate from safety thresholds, thereby ensuring proactive oversight of the industrial environment.

AnalogLink Sentinel: AIoT System Architecture

A the Complete, non-invasive Industrial pressure monitoring Solution

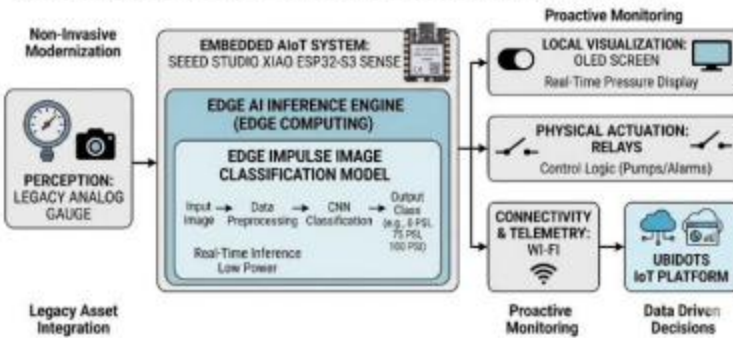


Fig 1. Block diagram and system architecture of AnalogLink Sentinel.

III. IMPLEMENTATION

The implementation process was structured as a seamless pipeline, spanning from data acquisition to local edge inference. Initially, a robust dataset was constructed by capturing pressure gauge imagery under diverse lighting conditions and angles to ensure environmental resilience. These samples were categorized into six operational classes: 0, 30, 60, 90, 120, and 150 psi. To comply with the stringent hardware constraints of the ESP32-S3 microcontroller, the images underwent preprocessing within the Edge Impulse platform. This stage involved resizing the images to 96x96 pixels and optimizing the color space to minimize computational overhead while maintaining feature integrity for the model.

Leveraging this dataset, a highly optimized image classification model was designed and trained specifically for computer vision tasks on microcontrollers. To ensure efficient, real-time execution on resource-constrained hardware, the model was quantized into an 8-bit integer (int8) format and exported as a native Arduino library. This

optimization drastically reduced RAM and Flash memory overhead (achieving up to 17% less RAM and 12% less ROM usage) without compromising the system's predictive performance. The integral stages constituting the AnalogLink Sentinel workflow are sequentially illustrated in Fig. 2.

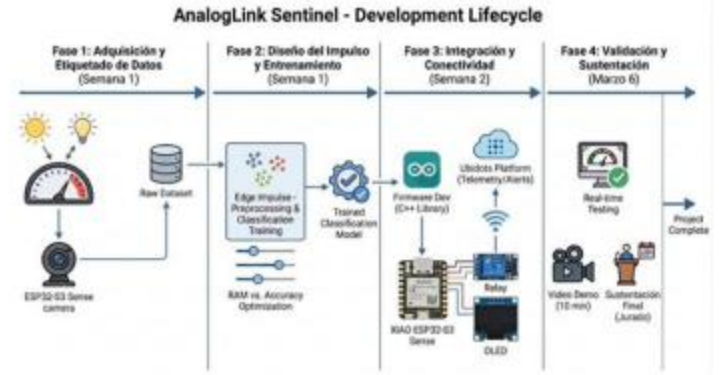


Fig 2. Workflow diagram of AnalogLink Sentinel.

IV. RESULTADOS

Recolección de datos: Construcción de un dataset balanceado mediante la captura de imágenes con el XIAO ESP32-S3 Sense, integrando variaciones de iluminación para simular ruido ambiental industrial y asegurar la generalización del modelo. La recolección del dataset se dio tomando 150 imágenes por cada clase definida (6 clases, desde 0 psi a 150 psi)

Data Acquisition: Construction of a balanced dataset through image capture utilizing the Seeed Studio XIAO ESP32-S3 Sense, incorporating lighting variations to simulate industrial environmental noise and ensure robust model generalization. The dataset acquisition process involved collecting 150 images for each of the six defined classes (ranging from 0 psi to 150 psi) to ensure categorical balance and representation.



Fig 3. Collected data distribution and training-test split for AnalogLink Sentinel.

Model Processing and Training: Preprocessing block configuration (including normalization and resizing) was tailored to both the visual characteristics of the dataset and the stringent hardware constraints of the ESP32-S3 Sense. Image classification model training was executed on the Edge Impulse platform, where hyperparameters such as training cycles (30) and learning rate (0.0005) were optimized to achieve a precise balance between diagnostic accuracy and the computational resource footprint



(RAM/Flash) of the embedded system. This phase leveraged the EON™ Compiler and int8 quantization to minimize memory overhead, reducing RAM usage by 17% and ROM by 12% compared to unoptimized versions.



Fig 4. Collected data distribution and train-test split for AnalogLink Sentinel.

Testing and Deployment: Following the training phase, the model was evaluated using unseen test data, achieving an overall classification accuracy of 98.38%. To ensure compatibility with the target hardware's limitations, the model underwent 8-bit quantization (int8) and was compiled utilizing the EON™ Compiler, which optimizes memory usage and execution efficiency for embedded systems. This optimization resulted in a model requiring only 182.8 KB of RAM, with an inference latency of 846 ms. Finally, the model was exported as a C++ library for the Arduino environment, enabling seamless integration and real-time operation on the ESP32-S3 Sense microcontroller.



Fig 5. Model Validation and Testing Results.

Firmware Integration and Connectivity: Firmware development was executed using the Arduino/C++ framework to integrate the Edge Impulse inference library into the Seeed Studio XIAO ESP32-S3 Sense hardware. The architecture leverages Wi-Fi connectivity and the HTTP

protocol for real-time telemetry transmission to the Ubidots IoT platform, ensuring continuous data logging and remote monitoring. Simultaneously, the system provides local feedback via an integrated OLED display for instantaneous pressure status visualization. A critical component of the firmware is the signal processing logic, which manages the treatment of acquired data and determines the operational pressure state based on confidence intervals, thereby optimizing the reliability of autonomous actuation and reporting.

```
if (mejor_probabilidad > 0.60) {  
  // Extraemos solo los digitos  
  int valor_numerico = extraerNumero(presion_detectada);  
  
  // Filtramos estrictamente por las 6 clases deseadas  
  if (valor_numerico == 0 || valor_numerico == 30 || valor_numerico == 60 ||  
      valor_numerico == 90 || valor_numerico == 120 || valor_numerico == 150) {  
  
    oled_line(2, String("> ") + String(valor_numerico) + " PSI ");  
    oled_line(4, String("Certeza: ") + String(mejor_probabilidad * 100, 1) + "%");  
  
    if (valor_numerico >= 120) {  
      oled_line(6, "!! PELIGRO !!");  
    } else if (valor_numerico >= 90) {  
      oled_line(6, "! PRECAUCION !");  
    } else {  
      oled_line(6, "Estado: NORMAL");  
    }  
  }  
}
```

Fig 6. Arduino/C++-based firmware developed for on-device deployment and real-time inference.

Final Validation: Following the deployment and verification phase, the system's performance was rigorously evaluated within the embedded environment. Real-time detection accuracy was compared against the metrics obtained during the validation stage, yielding highly consistent results for each established pressure class (0 to 150 psi). Nevertheless, minor discrepancies and operational latencies—specifically an inference time of approximately 846 ms—were observed during on-device execution.



Fig 7. Real-time data visualization and historical logging on the Ubidots dashboard.

IV. CONCLUSIONS AND FUTURE WORK

The development and implementation of AnalogLink Sentinel successfully demonstrated the technical feasibility of executing computer vision models directly on low-cost microcontrollers for the non-invasive digitization of analog instrumentation. Experimental results confirmed that the image classification approach is highly effective for discretizing pressure readings. Utilizing a balanced dataset of 150 images for each of the six operational classes, the model achieved a remarkable testing accuracy of 98.38%. A



critical factor for this performance level was the 8-bit integer (int8) quantization and the integration of the EON™ compiler. This neural network compression technique reduced RAM and Flash memory usage by 17% and 12%, respectively, enabling the vision model to operate seamlessly within the stringent hardware constraints of the Seeed Studio XIAO ESP32-S3 Sense.

Furthermore, the robustness of Edge AI inference was evidenced by the preprocessing stage, where resizing imagery to 96x96 pixels and adjusting the color space sufficed for extracting the needle's geometric features without requiring external processing. Validation tests confirmed that the system evaluates confidence intervals locally and in real-time, maintaining total independence from network status. However, comprehensive validation also revealed minor discrepancies and delays during telemetry transmission to the Ubidots platform. This behavior stems from a current firmware architectural bottleneck: the microcontroller faces significant computational overhead when concurrently managing video frame capture, tensor inference, and data transmission via the HTTP protocol, which is relatively resource-heavy for such operations. In summary, the system is consolidated as a highly reliable and functional local prototype for autonomous gauge status recognition.

Based on these findings, several opportunities for improvement are proposed to optimize system performance for future industrial deployment:

Firmware Restructuring: To resolve connectivity latencies, it is recommended to implement a Real-Time Operating System (RTOS), such as FreeRTOS, to parallelize tasks across the processor's dual cores, assigning image processing to one core and network stack management to the other.

Protocol Optimization: Replacing HTTP with the MQTT protocol is essential, as its lightweight, telemetry-oriented design will drastically reduce latency and processing load during data transmission.

Transition to Visual Regression: While range-based classification proved accurate, it limits readout resolution. Future work should explore visual regression architectures or optimized object detection algorithms, such as FOMO within Edge Impulse, to identify the needle's spatial

coordinates and calculate its exact angle for continuous numerical pressure reporting.

Environmental Robustness: To ensure high reliability under industrial plant dynamics, the training set should be expanded via data augmentation techniques that simulate extreme lighting variations, glass reflections, and debris accumulation, thereby strengthening the model's generalization capabilities against adverse operating conditions.

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